Assignment -11.1

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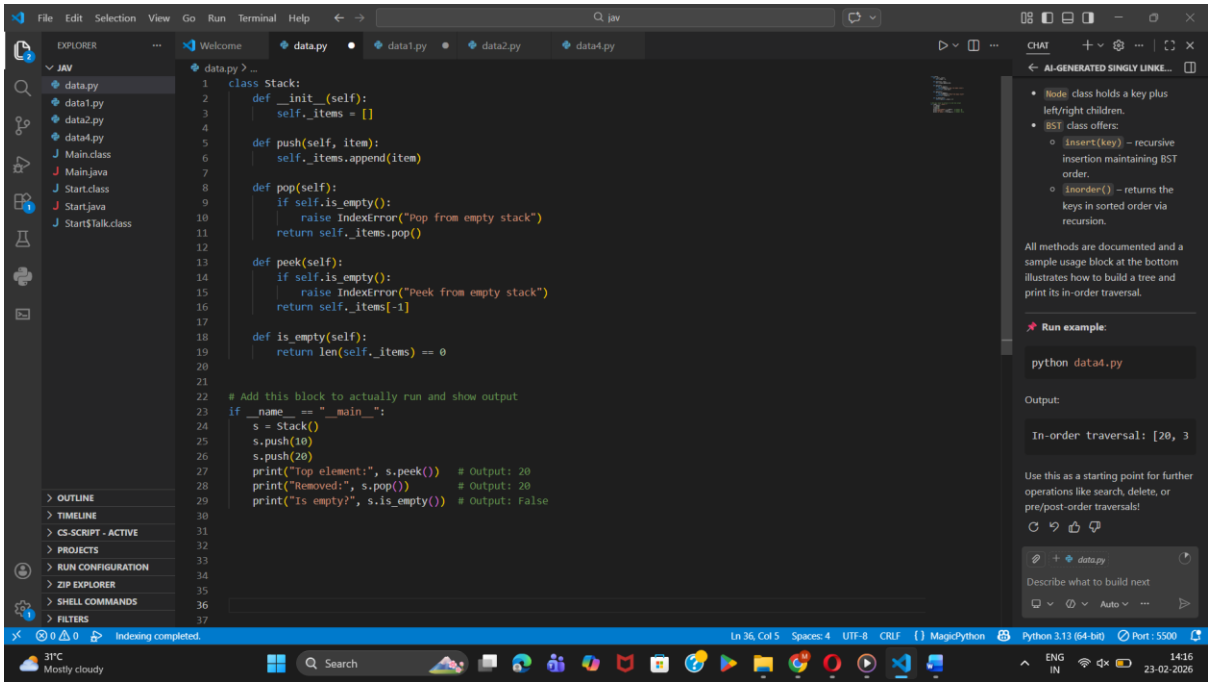
Batch-43



```python
class Queue:
    def __init__(self):
        self._items = []

    def enqueue(self, item):
        self._items.append(item)

    def dequeue(self):
        if self.is_empty():
            raise IndexError("Dequeue from empty queue")
        return self._items.pop(0)

    def peek(self):
        if self.is_empty():
            raise IndexError("Peek from empty queue")
        return self._items[0]

    def size(self):
        return len(self._items)

    def is_empty(self):
        return len(self._items) == 0


if __name__ == "__main__":
    q = Queue()
    q.enqueue(10)
    q.enqueue(20)
    q.enqueue(30)

    print("Front element:", q.peek())
```

```
Front element: 10
Removed: 10
Queue size: 2
Is empty? False
```

```python
"""Simple Python module demonstrating a singly linked list.

This file contains two classes, `Node` and `LinkedList`, along with
basic insert and display functionality, plus a small demonstration of
the classes when run as a script.
"""


class Node:
    """A node in a singly linked list.

    Attributes:
        data: The value held by the node.
        next: Reference to the next node in the list (or ``None``).
    """

    def __init__(self, data):
        self.data = data
        self.next = None


class LinkedList:
    """A simple implementation of a singly linked list."""

    def __init__(self):
        self.head = None

    def insert(self, data):
        """Append a new element containing ``data`` to the end of the list.

        Parameters
```

```
[Running] python -u "c:\Users\geeth\java1\jav\data2.py"
Linked list contents: [0, 10, 20, 30, 40]
After clearing: []
```

```python
class LinkedList:
    def insert(self, data):
        Parameters
        ----------
        data : any
            Value to store in the new node.
        """
        new_node = Node(data)
        if not self.head:
            self.head = new_node
            return
        current = self.head
        while current.next:
            current = current.next
        current.next = new_node

    def display(self):
        """Return a Python list of the values in the linked list."""
        values = []
        current = self.head
        while current:
            values.append(current.data)
            current = current.next
        return values

    def clear(self):
        """Remove all elements from the list."""
        self.head = None
```

```
[Running] python -u "c:\Users\geeth\java1\jav\data2.py"
Linked list contents: [0, 10, 20, 30, 40]
After clearing: []
```

```python
class LinkedList:
    def display(self):
        """Return a Python list of the values in the linked list."""
        values = []
        current = self.head
        while current:
            values.append(current.data)
            current = current.next
        return values

    def clear(self):
        """Remove all elements from the list."""
        self.head = None


if __name__ == "__main__":
    # quick demonstration
    ll = LinkedList()
    for i in range(5):
        ll.insert(i * 10)
    print("Linked list contents:", ll.display())
    ll.clear()
    print("After clearing:", ll.display())
```

```
[Running] python -u "c:\Users\geeth\java1\jav\data2.py"
Linked list contents: [0, 10, 20, 30, 40]
After clearing: []
```

```python
class Node:
    """A node in a binary search tree.

    Attributes:
        key: The value stored in the node (comparable).
        left: Left child node.
        right: Right child node.
    """

    def __init__(self, key):
        self.key = key
        self.left = None
        self.right = None


class BST:
    """Basic binary search tree supporting insertion and traversal."""

    def __init__(self):
        self.root = None

    def insert(self, key):
        """Insert `key` into the BST.

        This method is a simple wrapper that calls the recursive helper.
        """
        self.root = self._insert_recursive(self.root, key)

    def _insert_recursive(self, node, key):
        """Helper that returns the new subtree root after insertion."""
```

```
[Running] python -u "c:\Users\geeth\java1\jav\data4.py"
In-order traversal: [20, 30, 40, 50, 60, 70, 80]
```

```python
class BST:
    def _insert_recursive(self, node, key):
        return Node(key)
        if key < node.key:
            node.left = self._insert_recursive(node.left, key)
        elif key > node.key:
            node.right = self._insert_recursive(node.right, key)
        # if key == node.key, do nothing (ignore duplicates)
        return node

    def inorder(self):
        """Return a list of keys resulting from an in-order traversal."""
        return self._inorder_recursive(self.root)

    def _inorder_recursive(self, node):
        if node is None:
            return []
        return (
            self._inorder_recursive(node.left)
            + [node.key]
            + self._inorder_recursive(node.right)
        )


if __name__ == "__main__":
    tree = BST()
    for value in [50, 30, 70, 20, 40, 60, 80]:
        tree.insert(value)
    print("In-order traversal:", tree.inorder())
```

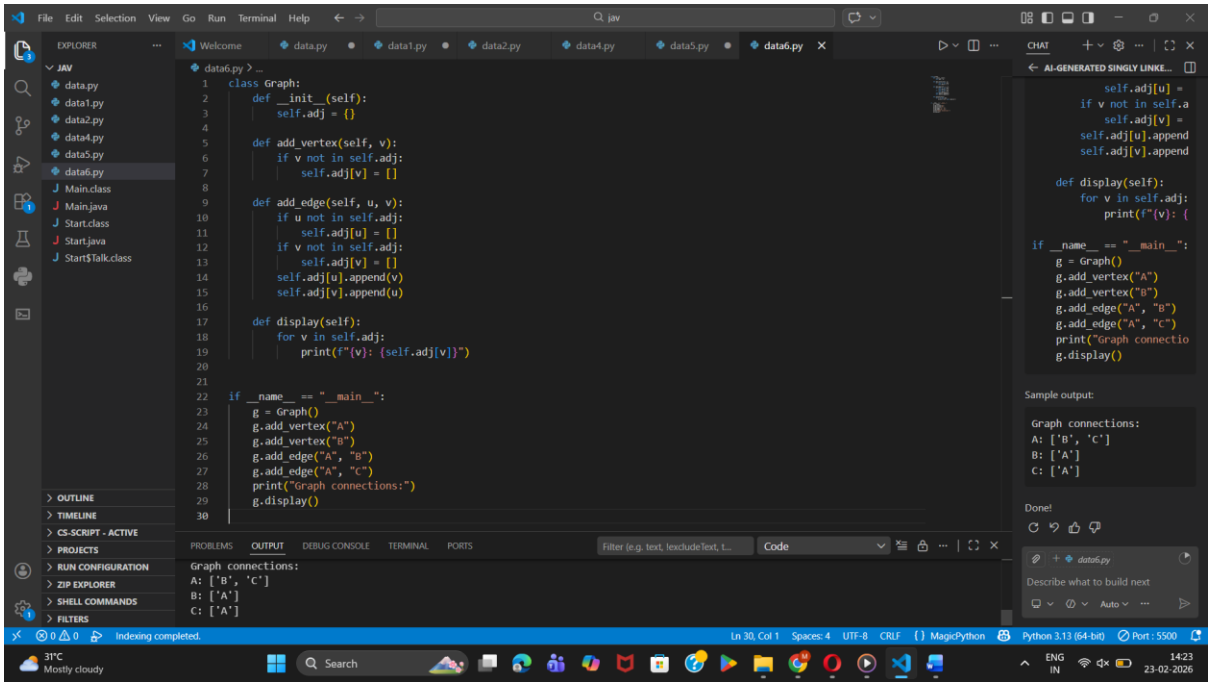
```
[Running] python -u "c:\Users\geeth\java1\jav\data4.py"
In-order traversal: [20, 30, 40, 50, 60, 70, 80]
```

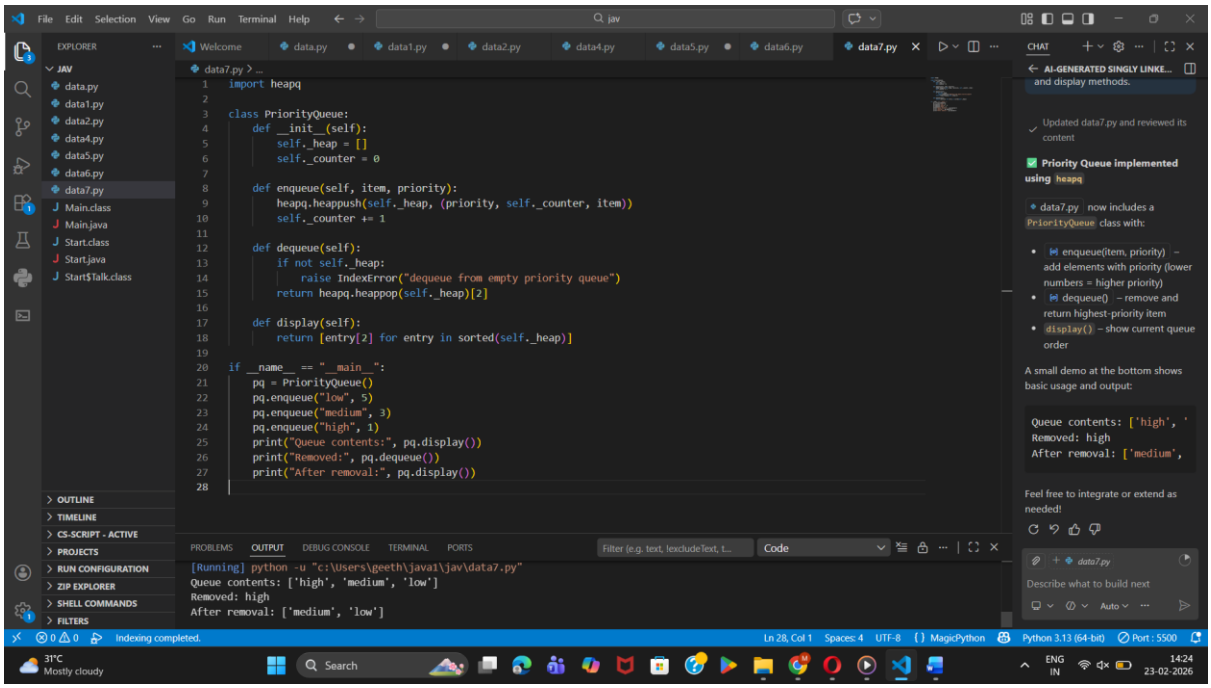
```python
class HashTable:
    """Basic hash table using chaining for collision resolution.

    Buckets are implemented as lists of (key, value) pairs.  The default
    number of buckets is small to force collisions when demonstrating
    behavior; adjust ``size`` as needed for real usage.
    """

    def __init__(self, size=10):
        # number of buckets
        self.size = size
        # create empty list for each bucket
        self.table = [[] for _ in range(size)]

    def _hash(self, key):
        """Compute an index for ``key`` into the underlying table."""
        return hash(key) % self.size

    def insert(self, key, value):
        """Insert or update a key/value pair in the table.

        If the key already exists in the bucket, its value is replaced.
        Otherwise the pair is appended to the chain.

        Parameters
        ----------
        key : hashable
            The key to insert.
        value : any
            The associated value to store.
```

```
[Running] python -u "c:\Users\geeth\java1\jav\data5.py"
cat -> meow
dog -> woof
dog after delete -> None
```

```python
    def search(self, key):
        """Return the value for ``key`` or ``None`` if not found."""
        idx = self._hash(key)
        for k, v in self.table[idx]:
            if k == key:
                return v
        return None

    def delete(self, key):
        """Remove ``key`` from the table if present.

        After deletion the bucket chain is adjusted accordingly.  If the
        key is not found, the method does nothing.
        """
        idx = self._hash(key)
        bucket = self.table[idx]
        for i, (k, _) in enumerate(bucket):
            if k == key:
                bucket.pop(i)
                return


if __name__ == "__main__":
    # simple demonstration
    ht = HashTable(size=5)
    ht.insert("cat", "meow")
    ht.insert("dog", "woof")
    ht.insert("bird", "tweet")
    print("cat ->", ht.search("cat"))
```

```
[Running] python -u "c:\Users\geeth\java1\jav\data5.py"
cat -> meow
dog -> woof
dog after delete -> None
```


data6.py > ...
class Graph:
def __init__(self):
self.adj = {}
def add_vertex(self, v):
if v not in self.adj:
self.adj[v] = []
def add_edge(self, u, v):
if u not in self.adj:
self.adj[u] = []
if v not in self.adj:
self.adj[v] = []
self.adj[u].append(v)
self.adj[v].append(u)
def display(self):
for v in self.adj:
print(f"{v}: {self.adj[v]}")
if __name__ == "__main__":
g = Graph()
g.add_vertex("A")
g.add_vertex("B")
g.add_edge("A", "B")
g.add_edge("A", "C")
print("Graph connections:")
g.display()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Graph connections:
A: ['B', 'C']
B: ['A']
C: ['A']
Sample output:
Graph connections:
A: ['B', 'C']
B: ['A']
C: ['A']
Done!


data7.py > ...
import heapq
class PriorityQueue:
def __init__(self):
self._heap = []
self._counter = 0
def enqueue(self, item, priority):
heapq.heappush(self._heap, (priority, self._counter, item))
self._counter += 1
def dequeue(self):
if not self._heap:
raise IndexError("dequeue from empty priority queue")
return heapq.heappop(self._heap)[2]
def display(self):
return [entry[2] for entry in sorted(self._heap)]
if __name__ == "__main__":
pq = PriorityQueue()
pq.enqueue("low", 5)
pq.enqueue("medium", 3)
pq.enqueue("high", 1)
print("Queue contents:", pq.display())
print("Removed:", pq.dequeue())
print("After removal:", pq.display())
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
[Running] python -u "c:\Users\geeth\java1\jav\data7.py"
Queue contents: ['high', 'medium', 'low']
Removed: high
After removal: ['medium', 'low']
Priority Queue implemented using heapq
Queue contents: ['high', '
Removed: high
After removal: ['medium',
Feel free to integrate or extend as needed!

```python
class DequeDS:

    def __init__(self):
        self._dq = deque()

    def insert_front(self, item):
        """Add an item to the front of the deque."""
        self._dq.appendleft(item)

    def insert_rear(self, item):
        """Add an item to the rear of the deque."""
        self._dq.append(item)

    def remove_front(self):
        """Remove and return an item from the front."""
        return self._dq.popleft()

    def remove_rear(self):
        """Remove and return an item from the rear."""
        return self._dq.pop()


if __name__ == "__main__":
    dq = DequeDS()
    dq.insert_rear(10)
    dq.insert_front(5)
    dq.insert_rear(15)
    print("Deque contents:", list(dq._dq))
    print("Removed front:", dq.remove_front())
    print("Removed rear:", dq.remove_rear())
```

```
[Running] python -u "c:\Users\geeth\java1\jav\data8.py"
Deque contents: [5, 10, 15]
Removed front: 5
Removed rear: 15
```

```python
import heapq
from typing import Any, List, Tuple


class PriorityQueue:
    def __init__(self) -> None:
        self._heap: List[Tuple[int, int, Any]] = []
        self._counter: int = 0

    def enqueue(self, item: Any, priority: int) -> None:
        heapq.heappush(self._heap, (priority, self._counter, item))
        self._counter += 1

    def dequeue(self) -> Any:
        if not self._heap:
            raise IndexError("dequeue from empty priority queue")
        return heapq.heappop(self._heap)[2]

    def peek(self) -> Any:
        if not self._heap:
            return None
        return self._heap[0][2]

    def display(self) -> List[Any]:
        return [entry[2] for entry in sorted(self._heap)]


def main() -> None:
    scheduler = PriorityQueue()
```

```
upcoming tasks: ['fix critical bug', 'deploy release', 'write report', 'check email']
executing: fix critical bug
executing: deploy release
executing: write report
executing: check email
```

```python
class PriorityQueue:
        return self._heap[0][2]

    def display(self) -> List[Any]:
        return [entry[2] for entry in sorted(self._heap)]


def main() -> None:
    scheduler = PriorityQueue()
    scheduler.enqueue("write report", 5)
    scheduler.enqueue("fix critical bug", 1)
    scheduler.enqueue("check email", 10)
    scheduler.enqueue("deploy release", 2)

    print("upcoming tasks:", scheduler.display())
    while True:
        try:
            task = scheduler.dequeue()
        except IndexError:
            break
        print("executing:", task)


if __name__ == "__main__":
    main()
```

```
upcoming tasks: ['fix critical bug', 'deploy release', 'write report', 'check email']
executing: fix critical bug
executing: deploy release
executing: write report
executing: check email
```